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clc; clear;

% Given cost coefficients
a = [500 400 200];
b = [5.3 5.5 5.8];
c = [0.004 0.006 0.009];

Pd = 800; % Total demand in MW
lambda = 6.0; % Initial guess for lambda
tol = 0.0001; % Convergence tolerance
dP = 1; % Initialize power mismatch

while abs(dP) > tol
    % Calculate power generated by each generator
    P1 = (lambda - b(1)) / (2 * c(1));
    P2 = (lambda - b(2)) / (2 * c(2));
    P3 = (lambda - b(3)) / (2 * c(3));
    % Total generation
    Pg = P1 + P2 + P3;
    % Power mismatch
    dP = Pd - Pg;
    % Update lambda using gradient method
    dLam = dP / (1/(2*c(1)) + 1/(2*c(2)) + 1/(2*c(3)));
    lambda = lambda + dLam;
end

% Display results
fprintf('\nEconomic Dispatch Results with No Limits \n');
fprintf('P1 = %.3f MW\n', P1);
fprintf('P2 = %.3f MW\n', P2);
fprintf('P3 = %.3f MW\n', P3);
fprintf('Total Generation = %.3f MW\n', Pg);
fprintf('Lambda = %.4f\n', lambda);

% Calculate total cost
C1 = a(1) + b(1)*P1 + c(1)*P1^2;
C2 = a(2) + b(2)*P2 + c(2)*P2^2;
C3 = a(3) + b(3)*P3 + c(3)*P3^2;
TotalCost = C1 + C2 + C3;
fprintf('Total Cost = %.3f Rs/hr\n', TotalCost);

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Economic Dispatch Results with No Limits
P1 = 400.000 MW
P2 = 250.000 MW
P3 = 150.000 MW
Total Generation = 800.000 MW
Lambda = 8.5000
Total Cost = 6682.500 Rs/hr

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